

MongoDB CRUD Practice – 15 Scenario-Based Problems

Assume each problem has its own collection unless specified otherwise.

1. College Student Management System

A college maintains student records in a **students** collection with fields such as:

studentId, name, branch, year, cgpa, city, skills

Perform the following:

1. Insert **5 student documents** .
 2. Display all students.
 3. Display students belonging to the **CSE branch** .
 4. Find students having a **CGPA greater than 8.0** .
 5. Update the city of a particular student.
 6. Delete a student who has withdrawn from the college.
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2. Online Book Store

An online bookstore stores books in a **books** collection containing:

bookId, title, author, category, price, stock

Perform the following:

1. Add **8 books** to the collection.
 2. Display all books belonging to the **Programming** category.
 3. Find books costing less than ₹500.
 4. Increase the price of a particular book.
 5. Update the stock of a book after receiving new inventory.
 6. Remove a book that is no longer available for sale.
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3. Employee Management System

A company stores employee information in an **employees** collection:

empId, name, department, designation, salary, experience, city

Perform the following:

1. Insert **10 employee records** .
 2. Display all employees working in the **IT department** .
 3. Find employees whose salary is greater than ₹60,000.
 4. Update the designation of an employee after promotion.
 5. Increase the salary of an employee by ₹5,000.
 6. Delete an employee who has left the organization.
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4. E-Commerce Product Catalog

An e-commerce website maintains products in a **products** collection:

productId, name, category, brand, price, quantity, rating

Perform the following:

1. Insert at least **10 products** .
 2. Display all products belonging to the **Electronics** category.
 3. Find products with a rating of **4 or above** .
 4. Update the price of a particular product.
 5. Increase the quantity of a product after restocking.
 6. Delete a discontinued product.
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5. Hospital Patient Records

A hospital maintains patient information in a **patients** collection:

patientId, name, age, gender, disease, doctor, roomNo

Perform the following:

1. Insert **8 patient records** .
 2. Display all patients suffering from **Diabetes** .
 3. Find patients whose age is above 60.
 4. Change the room number of a patient.
 5. Update the doctor assigned to a patient.
 6. Delete the record of a patient who has been discharged.
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6. Movie Streaming Platform

A streaming platform stores movies in a **movies** collection:

movieId, title, genre, language, year, rating, duration

Perform the following:

1. Insert **10 movie documents** .
 2. Display all movies in the **Action** genre.
 3. Find movies released after 2020.
 4. Find movies having a rating greater than 8.
 5. Update the rating of a movie.
 6. Delete a movie that has been removed from the platform.
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7. Food Delivery Application

A food delivery application maintains restaurant information in a **restaurants** collection:

restaurantId, name, cuisine, city, rating, deliveryTime, isOpen

Perform the following:

1. Add **8 restaurants** .
2. Display restaurants serving **Indian cuisine** .
3. Find restaurants with a rating above 4.0.
4. Update the delivery time of a restaurant.
5. Change **isOpen** when a restaurant closes.
6. Delete a restaurant that permanently shuts down.

8. College Course Registration

A university maintains courses in a **courses** collection:

courseId, courseName, department, credits, faculty, seats

Perform the following:

1. Insert **10 courses** .
2. Display all courses offered by the **CSE department** .
3. Find courses having **4 credits** .
4. Update the faculty assigned to a course.
5. Reduce the number of available seats after student registration.
6. Delete a course that is no longer offered.

9. Banking Customer Management

A bank stores customer information in a **customers** collection:

customerId, name, accountType, balance, city, status

Perform the following:

1. Insert **10 customer records** .
2. Display customers having a **Savings** account.
3. Find customers whose balance is greater than ₹1,00,000.
4. Update the balance of a customer after a deposit.
5. Change the account status from **Active** to **Inactive**.
6. Delete a customer account that has been permanently closed.

10. Vehicle Rental System

A vehicle rental company maintains vehicles in a **vehicles** collection:

vehicleId, type, brand, model, year, rentPerDay, availability

Perform the following:

1. Insert **10 vehicles** .
 2. Display all available cars.
 3. Find vehicles having a rental price below ₹2,000 per day.
 4. Change a vehicle's availability after it is rented.
 5. Update the rental price of a vehicle.
 6. Delete a vehicle that has been permanently removed from the fleet.
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11. Employee Attendance System

A company stores attendance information in an **attendance** collection:

attendanceId, empId, employeeName, date, status, workingHours

Perform the following:

1. Insert attendance records for multiple employees.
 2. Display all employees marked **Absent** .
 3. Find employees who worked more than 8 hours.
 4. Correct the attendance status of an employee mistakenly marked absent.
 5. Update working hours for a particular attendance record.
 6. Delete an attendance record entered incorrectly.
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12. Online Learning Platform

An online learning platform maintains courses in a **courses** collection:

courseId, title, instructor, category, duration, fee, studentsEnrolled

Perform the following:

1. Insert **10 online courses** .
2. Display all courses in the **Data Science** category.
3. Find courses costing less than ₹5,000.
4. Update the course fee.
5. Increase **studentsEnrolled** after new registrations.

6. Delete a course that is no longer available.

13. Hotel Reservation System

A hotel stores room information in a **rooms** collection:

roomNo, type, price, floor, capacity, status

Perform the following:

1. Insert records for **10 rooms** .
 2. Display all available rooms.
 3. Find rooms with a capacity of at least 3 people.
 4. Update the status of a room after booking.
 5. Change the price of a room after a pricing revision.
 6. Delete a room that is permanently unavailable.
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14. Internship Management System

A college maintains internship opportunities in an **internships** collection:

internshipId, company, domain, location, stipend, duration, openings

Perform the following:

1. Insert **10 internship opportunities** .
 2. Display internships related to **Artificial Intelligence** .
 3. Find internships offering a stipend greater than ₹20,000.
 4. Update the number of openings after students are selected.
 5. Update the location of an internship.
 6. Delete an internship after its application deadline has passed.
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15. Student Project Management System

A department maintains student project information in a **projects** collection:

projectId, title, domain, students, guide, status, year

Perform the following:

1. Insert at least **10 project records** .
 2. Display all projects belonging to the **AI/ML domain** .
 3. Find all projects whose status is **Completed**.
 4. Update the project status from **In Progress** to **Completed**.
 5. Change the faculty guide assigned to a project.
 6. Delete a project that has been cancelled.
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